

Assignment 02: Exploratory Data Analysis (EDA)

Case Study: The "Titanic" of Agriculture

Module: Introduction to Machine Learning

Instructions

- **Type:** Analytical Essay and Visualization Design.
- **Goal:** This assignment evaluates your ability to "listen" to data before modeling. You must demonstrate the ability to characterize data distributions, identify anomalies, and formulate hypotheses based on visual evidence.
- **Context:** You are not just calculating statistics; you are investigating a phenomenon.

Case Study: Project "Green Hope" (Crop Yield Crisis)

You are a consultant for an agricultural consortium in a developing nation. They have collected historical data from 5,000 small farms to predict **Crop Yield** (y , tons/hectare). The dataset $\mathcal{D}$ contains the following features:

- x_1 : **Rainfall** (mm/year) - Continuous.
- x_2 : **Fertilizer Usage** (kg/hectare) - Continuous.
- x_3 : **Soil Type** (Clay, Loam, Sandy) - Categorical.
- x_4 : **Pest Infestation Level** (0 to 10 scale) - Ordinal.

The Problem: Initial linear models failed miserably. The client believes "more fertilizer always means more crop," but your preliminary look suggests something weird is happening in the data. They need your EDA expertise to explain why simple assumptions are failing.

Part 1: Data Characterization & Distribution (Cognitive Analysis)

Before checking correlations, we must understand the "shape" of our variables.

Q1. The Gaussian vs. The Skewed: Most natural phenomena (like Rainfall) tend to follow a Normal (Bell-shaped) Distribution. However, human-driven factors (like Fertilizer Usage) often follow a Right-Skewed distribution (most farmers use a little, a few rich farmers use a lot).

- Sketch two hypothetical histograms: one for x_1 (Rainfall) assuming a Gaussian distribution, and one for x_2 (Fertilizer) assuming a Right-Skewed distribution. Label axes clearly.
- Explain why applying a "Mean" summary statistic to the skewed fertilizer data might be misleading compared to using the "Median."

Q2. Categorical Encoding: The feature x_3 (Soil Type) is text ("Clay", "Loam"). The machine learning model requires numerical input $\mathbf{X} \in \mathbb{R}^{N \times D}$.

- Propose a method to convert this categorical feature into numbers (e.g., One-Hot Encoding vs. Label Encoding).
- Explain why mapping "Clay" = 1, "Loam" = 2, "Sandy" = 3 might be dangerous for a Linear Model. (Hint: Does $2 \times \text{Clay} = \text{Loam}$?)

Part 2: Visualizing Relationships (Pattern Recognition)

The client claims: *"Just plot Fertilizer (x_2) vs Yield (y). It should be a straight line up."*

Q3. The Non-Linear Reality: Biologically, plants die if over-fertilized (toxicity).

- Sketch a Scatter Plot of x_2 (x-axis) vs. y (y-axis) that demonstrates the law of diminishing returns and eventual toxicity (an inverted U-shape).
- Mark the region on your plot where a simple Linear Model ($y = wx + b$) would make catastrophic errors.

Q4. Multivariate Interaction: You suspect that Rainfall (x_1) only helps if the Soil Type (x_3) is right.

- Describe how you would visualize this 3-variable relationship (y vs x_1 grouped by x_3). Would you use color? Separate subplots? Explain your design choice.

Part 3: Data Cleaning & Stochasticity (Critical Thinking)

Upon closer inspection, you find "dirty" data.

Q5. Missing Values (NaNs): 5% of the farms have missing values for x_4 (Pest Infestation).

- Scenario A: The missing data is from a region where sensors broke randomly.
- Scenario B: The missing data is from farms that yielded 0 crops because they were abandoned due to extreme pests.
- **Task:** How should you handle the missing values in Scenario A vs. Scenario B? (Imputation vs. Flagging). Justify your answer.

Q6. Outliers or Realities?: You find a data point: *Rainfall = 2000mm, Yield = 0*.

- Is this an "error" (noise ϵ) that should be removed, or a "signal" (flood) that the model must learn?
- How does your decision affect the model's ability to generalize to future extreme weather events?

Part 4: Conclusion (HOTS)

Q7. The EDA Report: Based on your analysis above (Skewed fertilizer, non-linear toxicity, soil interactions), refute the client's initial request for a simple Linear Model.

- Write a concise paragraph explaining why a model of the form $y = w_1x_1 + w_2x_2 + \dots + b$ is structurally incapable of capturing the agricultural reality you discovered.
- Suggest one feature transformation (e.g., x_2^2) that might fix the "toxicity" issue in a linear framework.